



Two Scripts, Both Repeating

Quick facilitation notes & answer key

Goal

Students see that two scripts run at the same time (concurrent), each with its own repeat loop and count. They count both, change one independently, write a third, and compare counts.

Curriculum (Ontario Math 2020, Strand C3)

C3.1 — solve problems and create computational representations of mathematical situations by writing and executing code, including code that involves sequential, concurrent, and repeating events.

C3.2 — read and alter existing code, including code that involves sequential, concurrent, and repeating events, and describe how changes to the code affect the outcomes.

Materials

A block-coding app (Scratch-style) to project — show two green-flag scripts side by side; the printed worksheet + a pencil per student.

Run it (about 30 min)

1. Hook: clap AND march together — two things at once is concurrent.
2. Project the two green-flag scripts; click the flag, watch both loops run together.
3. Students do Exercises 1–4 (Ex 4 is the stretch), solo or in pairs.
4. Share answers; land the big idea: each script has its own loop and count.

Answer key (checked by the run-gate)

Ex 1 — 4 moves and 4 sounds.

Ex 2 — 4 moves and 6 sounds; changing B does NOT change A (independent).

Ex 3 — repeat 3.

Ex 4 (challenge) — Script A finishes more: 5 vs 2, so 3 more repeats.

Watch for & differentiate

Common slip: thinking one script must finish before the other starts — they run at the same time.

Support: trace each script with a different finger at the same time.

Extend: ask which finishes first if A is repeat 2 and B is repeat 6 (A).

Success looks like

The student counts both loops (Ex 1), reasons that changing one script leaves the other unaffected (Ex 2), writes a third concurrent loop (Ex 3), and compares two counts (Ex 4).